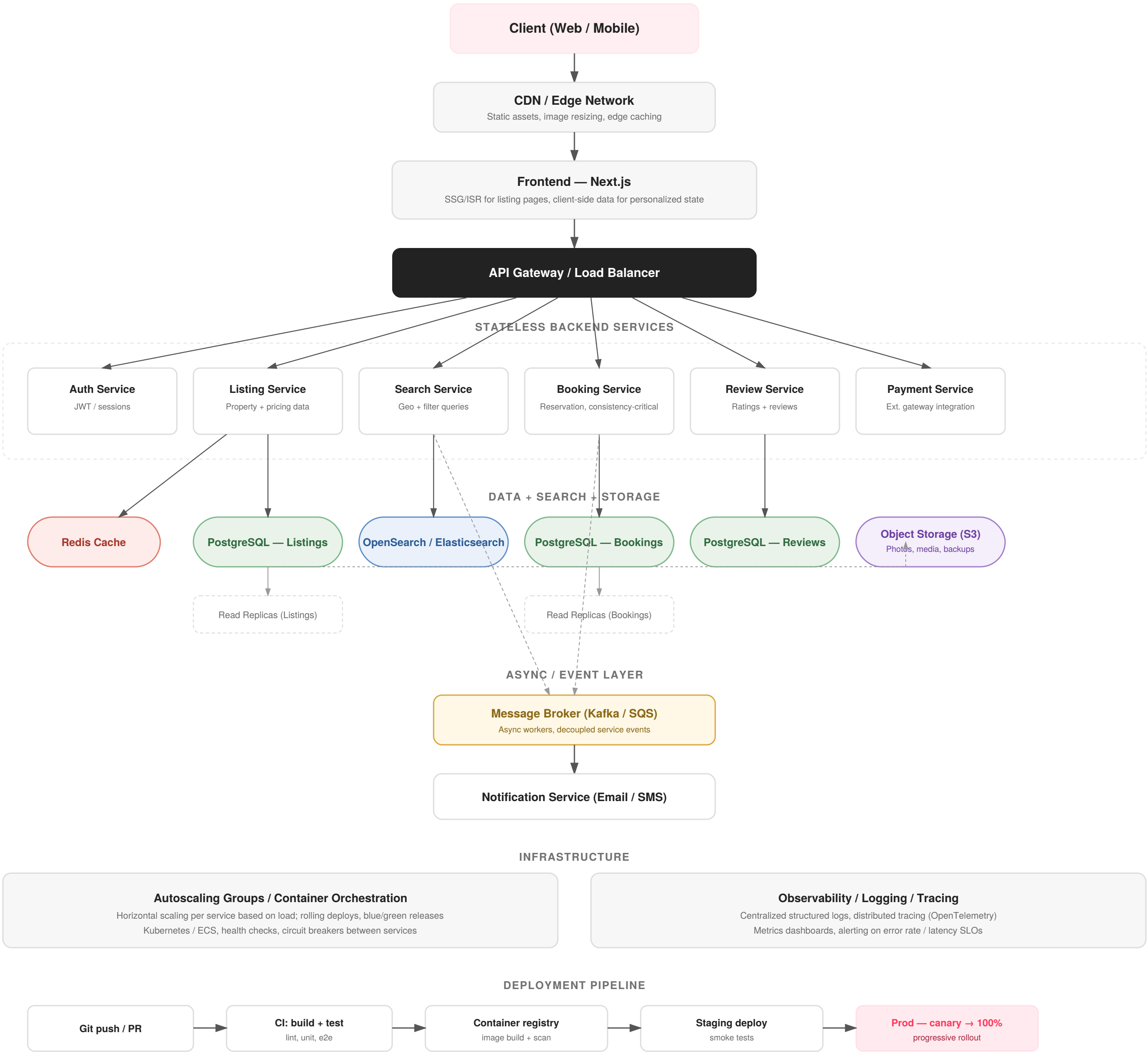


Production Architecture — Vacation Rental Marketplace

Scaling strategy: frontend delivery, backend services, storage, search, deployment



Scaling strategy summary

Frontend:

Listing pages pre-rendered (SSG/ISR) and served from the CDN edge — most traffic never reaches origin. Personalized data is fetched client-side.

Backend:

Stateless services behind the gateway scale horizontally and independently — a booking traffic spike doesn't require scaling search, and vice versa.

Storage:

Read-heavy listing data offloaded to Redis and read replicas. Booking writes stay on a smaller, strongly-consistent primary to avoid double-booking.

Bottleneck at current scale:

Booking Service — it's the only strongly-consistent write path (no double-booking), so it scales via read replicas + row-level locking on availability, not just adding more stateless instances.

Search:

Denormalized listing index in OpenSearch, updated async via the message broker on listing/price/availability changes — keeps writes off the read path.

Deployment:

Canary releases per service with automated rollback on error-rate/latency regression; blue/green for the frontend to avoid asset/version mismatches.

Resilience:

Circuit breakers between services; payment/booking failures degrade gracefully rather than cascading into search/browse traffic.